

UR5e Control Analysis

Part 1B-3 — Discrete LQR Controller Design and Nominal Benchmark

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Abstract

This report documents the design, diagnosis, reformulation, and nominal nonlinear benchmark of a discrete linear-quadratic regulator (LQR) for the validated UR5e rigid-body model established in Part 1A. The study intentionally retained unsuccessful intermediate designs because they exposed important differences between local linear stability, sampled-data implementation, and nonlinear trajectory tracking.

The initial controller was designed from a continuous-time linearization about the benchmark initial equilibrium. Although the linearized closed loop was controllable and mathematically stable, it contained an extremely fast mode of approximately $-1.23 \times 10^4 \text{ s}^{-1}$. When the resulting fixed-equilibrium controller was applied to the nonlinear 4 s trajectory with a 2 ms control period, performance collapsed: overall joint RMSE reached 435.19° and Joint 6 saturated at the adopted 28 N m limit.

I therefore reformulated the design directly in discrete time using exact zero-order-hold discretization and the discrete algebraic Riccati equation. Bryson-style state and input normalization was introduced, followed by a global control-penalty study. Within the predefined nonlinear comparison range, $\rho_u = 100$ provided the best tracking among the tested candidates. However, the resulting fixed-gain LQR still produced 0.8105° overall joint RMSE and 19.474 mm TCP position RMSE when gravity feedforward was frozen at the initial equilibrium.

A controlled A/B experiment then isolated the dominant structural issue. Replacing fixed $g(q_0)$ with reference-dependent $g(q_{\text{ref}})$ while leaving the discrete gain, plant, trajectory, constraints, and sample period unchanged reduced overall joint RMSE by 98.09% to 0.015502° and TCP position RMSE by 97.85% to 0.4185 mm. No torque saturation occurred, and maximum torque utilization was 27.22%. The final Python implementation required 0.514 ms mean, 0.525 ms 99th-percentile, and 0.914 ms maximum controller computation time in the present benchmark environment.

The final controller is therefore a fixed-gain discrete LQR with reference-dependent gravity compensation. It provides a strong nominal benchmark while preserving a clear distinction between local optimal state feedback and nonlinear model-based feedforward compensation.

1 Scope and Benchmark

The nonlinear plant, kinematics, rigid-body dynamics, equilibrium analysis, and common trajectory benchmark were established and cross-validated in Part 1A. The purpose of this stage is not to claim a globally optimal controller, but to investigate how an LQR designed from a local linear model behaves when applied to the full nonlinear UR5e trajectory and to identify which design changes materially improve that behavior.

All nominal controller comparisons use the same conditions:

- 4.0 s quintic joint-space trajectory and 2 ms sample period (500 Hz);
- $q_0 = [0, -90, 90, -90, -90, 0]^\circ$;
- $q_f = [20, -60, 60, -70, -70, 20]^\circ$;

- $|\tau_{1:3}| \leq 150 \text{ N m}$ and $|\tau_{4:6}| \leq 28 \text{ N m}$;
- nonlinear rigid-body plant integrated by RK4 with zero-order-held control torque;
- nominal fixed-base arm with no external payload.

2 LQR Formulation

The nonlinear rigid-body dynamics are

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau. \quad (1)$$

For local LQR design, the state is

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, \quad (2)$$

and the nonlinear dynamics are written as

$$\dot{x} = f(x, \tau). \quad (3)$$

At the benchmark initial equilibrium,

$$x_{\text{eq}} = \begin{bmatrix} q_0 \\ 0 \end{bmatrix}, \quad \tau_{\text{eq}} = g(q_0), \quad (4)$$

the model was numerically linearized as

$$\delta\dot{x} = A\delta x + B\delta\tau. \quad (5)$$

The equilibrium linearization was fully controllable:

$$\text{rank } \mathcal{C} = 12, \quad (6)$$

where $\mathcal{C}$ is the controllability matrix.

For a quadratic objective,

$$J = \int_0^\infty \left(\delta x^\top Q \delta x + \delta \tau^\top R \delta \tau \right) dt, \quad (7)$$

the continuous controller has the familiar form

$$\delta\tau = -K\delta x. \quad (8)$$

For trajectory tracking, the state deviation is defined as

$$\delta x = \begin{bmatrix} q - q_{\text{ref}} \\ \dot{q} - \dot{q}_{\text{ref}} \end{bmatrix}. \quad (9)$$

3 Initial Continuous-Time Design

The first implementation used the continuous-time algebraic Riccati equation with

$$Q = \text{diag}(100, \dots, 100, 10, \dots, 10), \quad R = I_6. \quad (10)$$

The linearized open-loop system contained an unstable mode with real eigenvalue approximately

$$+5.093 \text{ s}^{-1}, \quad (11)$$

while the LQR-stabilized linear model had all closed-loop poles in the left half-plane. The largest real part was approximately -2.606 s^{-1} .

However, the closed-loop pole set also contained very fast modes, including approximately

$$-1.228 \times 10^4, \quad -749.5, \quad -258.8 \text{ s}^{-1}. \quad (12)$$

The fastest mode corresponds to a time scale of only about $81 \mu\text{s}$, far below the 2 ms controller sample period. This did not invalidate the continuous-time stability calculation, but it immediately raised a sampled-data implementation concern.

4 Failure of the Initial Nonlinear Trajectory Benchmark

The continuous fixed-equilibrium LQR was then applied to the full nonlinear plant using the common 2 ms benchmark. The feedforward torque was held at the initial equilibrium value,

$$\tau_{\text{ff}} = g(q_0). \quad (13)$$

The result was a clear failure rather than a marginal degradation. Overall joint RMSE reached

$$\boxed{435.1889^\circ}, \quad (14)$$

and Joint 6 diverged until it repeatedly reached the adopted 28 N m torque limit. Joint 6 RMSE alone exceeded 1065° .

The failure demonstrates that local continuous-time stability at one equilibrium is not sufficient evidence of successful nonlinear sampled trajectory tracking. Three issues were present simultaneously: the controller was designed in continuous time but executed at 2 ms intervals, the gain was fixed from a single equilibrium linearization, and the gravity feedforward remained frozen at the initial configuration while the robot moved through a substantially different trajectory.

Rather than tuning around this failed architecture, I reformulated the controller directly for the sampled implementation.

5 Discrete-Time Reformulation

The continuous linear model was discretized at

$$T_s = 0.002 \text{ s} \quad (15)$$

using exact zero-order hold:

$$\delta x_{k+1} = A_d \delta x_k + B_d \delta \tau_k. \quad (16)$$

The discrete quadratic objective is

$$J = \sum_{k=0}^{\infty} \left(\delta x_k^\top Q \delta x_k + \delta \tau_k^\top R \delta \tau_k \right), \quad (17)$$

and the gain is obtained from the discrete algebraic Riccati equation,

$$K = \left(R + B_d^\top P B_d \right)^{-1} B_d^\top P A_d. \quad (18)$$

The discrete closed-loop system is

$$A_{\text{cl}} = A_d - B_d K, \quad (19)$$

with stability assessed from

$$\rho(A_{\text{cl}}) < 1. \quad (20)$$

6 Bryson Normalization and Control-Penalty Study

6.1 Normalized weights

Because joint position, joint velocity, and torque have different physical units and scales, I replaced the arbitrary initial weighting with Bryson-style normalization. The selected design scales were

$$q_{\text{allow}} = 2^\circ, \quad \dot{q}_{\text{allow}} = 10^\circ/\text{s}. \quad (21)$$

The state weights were

$$Q_{ii} = \frac{1}{x_{i,\text{allow}}^2}, \quad (22)$$

and the input normalization used the adopted simulation torque limits:

$$R_{\text{Bryson},jj} = \frac{1}{\tau_{j,\text{allow}}^2}. \quad (23)$$

A global input penalty was then introduced:

$$R = \rho_u R_{\text{Bryson}}. \quad (24)$$

The “allowable” values are normalization and design scales, not hard state constraints.

6.2 Linear pole and gain sweep

I first examined

$$\rho_u \in \{1, 10, 100, 1000, 10000\}. \quad (25)$$

Table 1: Discrete LQR control-penalty sweep. Equivalent continuous poles are computed from $\log(z)/T_s$ for interpretation only.

ρ_u	Spectral radius	Fastest equiv. pole [s ⁻¹]	$\ K\ _2$	$\max K_{ij} $
1	0.9900498649	-7127.821	3294.154	2908.184
10	0.9900498602	-5976.534	1285.481	1202.867
100	0.9900498439	-4825.300	462.959	448.196
1000	0.9900497590	-3674.586	184.095	179.361
10000	0.9918873065	-2529.028	101.928	98.869

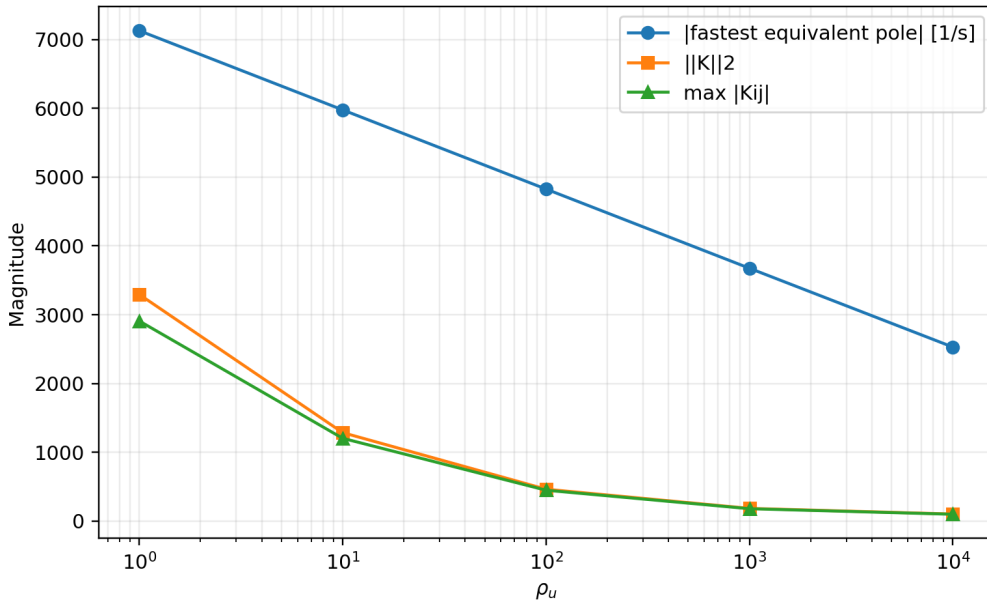


Figure 1: Effect of the global input penalty on gain magnitude and the fastest equivalent continuous pole. Increasing ρ_u reduces feedback gain and relaxes the fastest modes, while the dominant discrete spectral radius remains close to 0.99 over most of the tested range.

The sweep was not extended indefinitely merely to make the fastest mode slower. Linear pole location alone does not determine nonlinear trajectory performance, so representative candidates were taken into the common nonlinear benchmark.

6.3 Nonlinear candidate comparison

The candidates $\rho_u = 100$, 1000, and 10000 were evaluated on the nonlinear 4 s trajectory with the same fixed gravity feedforward $g(q_0)$.

Table 2: Nonlinear trajectory results for the discrete LQR control-penalty candidates.

ρ_u	Joint RMSE [deg]	TCP RMSE [mm]	TCP max [mm]	Ori. RMSE [deg]	Max util. [%]
100	0.810459	19.473746	33.601994	1.761316	28.344
1000	1.973029	54.664998	95.139900	4.142316	30.251
10000	4.085329	118.464330	206.604915	8.706249	32.942

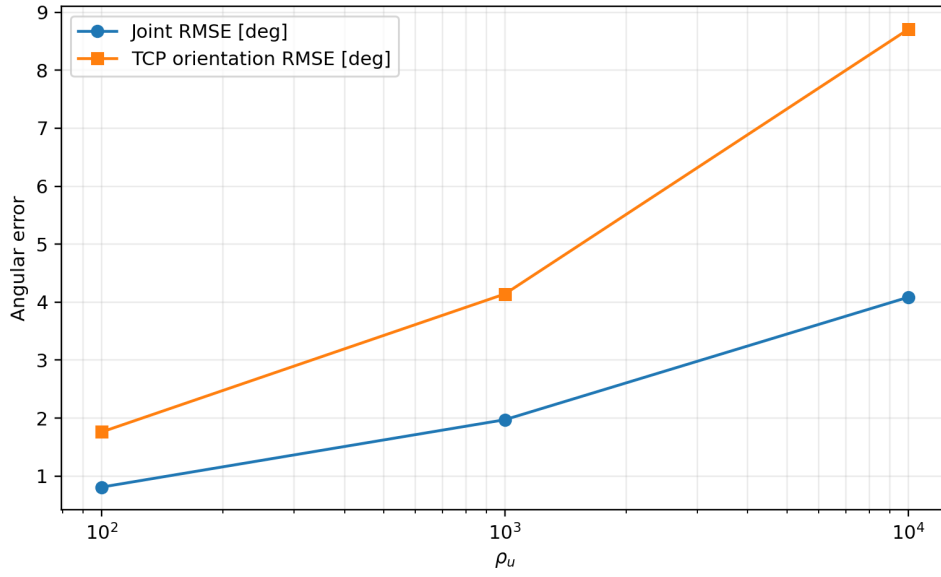


Figure 2: Angular tracking metrics for the nonlinear ρ_u comparison. Tracking degrades substantially as the control penalty increases over the tested range.

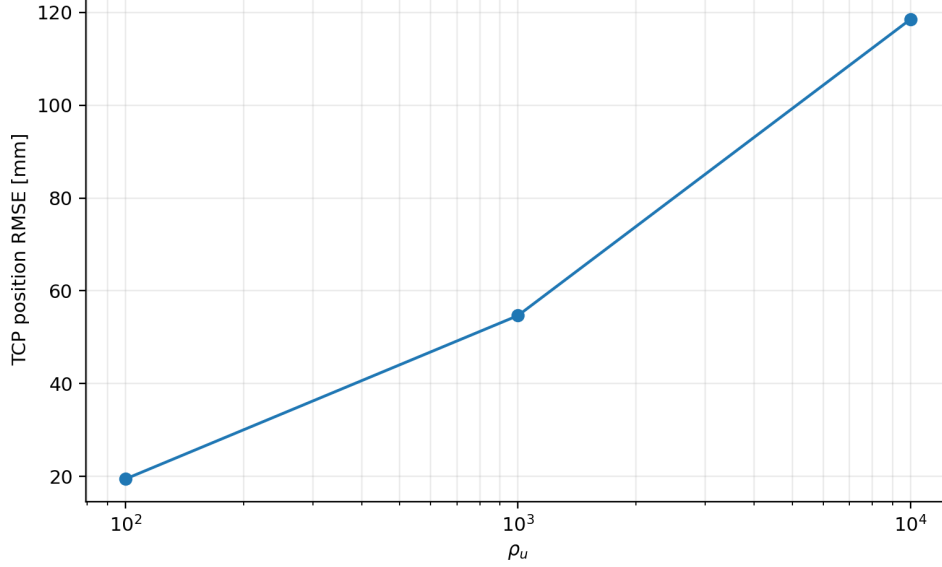


Figure 3: TCP position RMSE for the nonlinear ρ_u comparison. The degradation is not caused by saturation; no candidate saturated in this study.

Within this predefined candidate range, $\rho_u = 100$ provided the best nonlinear tracking and was selected for the next diagnostic step. This is a study-specific nominal selection, not a claim of a global mathematical optimum.

An initially counterintuitive result was that peak Joint 2 torque increased as ρ_u increased: approximately 42.52, 45.38, and 49.41 N m for $\rho_u = 100, 1000$, and 10000 , respectively. Increasing the input penalty therefore did not monotonically reduce the peak torque observed on the nonlinear trajectory. The weaker feedback allowed larger tracking deviations to develop, which subsequently required larger corrective torque. This illustrates why a Riccati weight should not be interpreted as a direct hard bound on realized peak input.

7 Diagnosis of Fixed-Equilibrium Gravity Feedforward

Even with the selected $\rho_u = 100$ discrete design, the nonlinear benchmark remained substantially worse than expected:

$$\text{Joint RMSE} = 0.810459^\circ, \quad (26)$$

$$\text{TCP position RMSE} = 19.473746 \text{ mm}. \quad (27)$$

The largest joint errors were concentrated in gravity-sensitive axes. Joint 2 RMSE was 1.681° and Joint 5 RMSE was 1.025° . No joint saturated.

This suggested that the problem was not insufficient modeled actuator authority. The controller was using

$$\tau = g(q_0) - K \begin{bmatrix} q - q_{\text{ref}} \\ \dot{q} - \dot{q}_{\text{ref}} \end{bmatrix}. \quad (28)$$

As the reference moved away from q_0 , the fixed term $g(q_0)$ became an increasingly inaccurate representation of the gravitational torque required along the trajectory. The LQR feedback was therefore responsible for both trajectory-error correction and compensation of a predictable configuration-dependent gravity mismatch.

8 Reference-Dependent Gravity Compensation

To isolate this hypothesis, I performed an A/B experiment in which only the feedforward term was changed. The discrete gain K , ρ_u , plant, trajectory, sample period, torque limits, and integration

method were held fixed.

The two controllers were

$$\tau_A = g(q_0) - K\delta x, \quad (29)$$

$$\tau_B = g(q_{\text{ref}}) - K\delta x. \quad (30)$$

Table 3: Controlled comparison of fixed and reference-dependent gravity feedforward.

Feedforward	Joint RMSE [deg]	TCP RMSE [mm]	TCP max [mm]	Ori. RMSE [deg]	Max util. [%]
Fixed $g(q_0)$	0.810459	19.473746	33.601994	1.761316	28.34
Reference $g(q_{\text{ref}})$	0.015502	0.418511	0.711982	0.037763	27.21

The reference-dependent gravity term reduced overall joint RMSE by

$$\boxed{98.09\%} \quad (31)$$

and TCP position RMSE by

$$\boxed{97.85\%}. \quad (32)$$

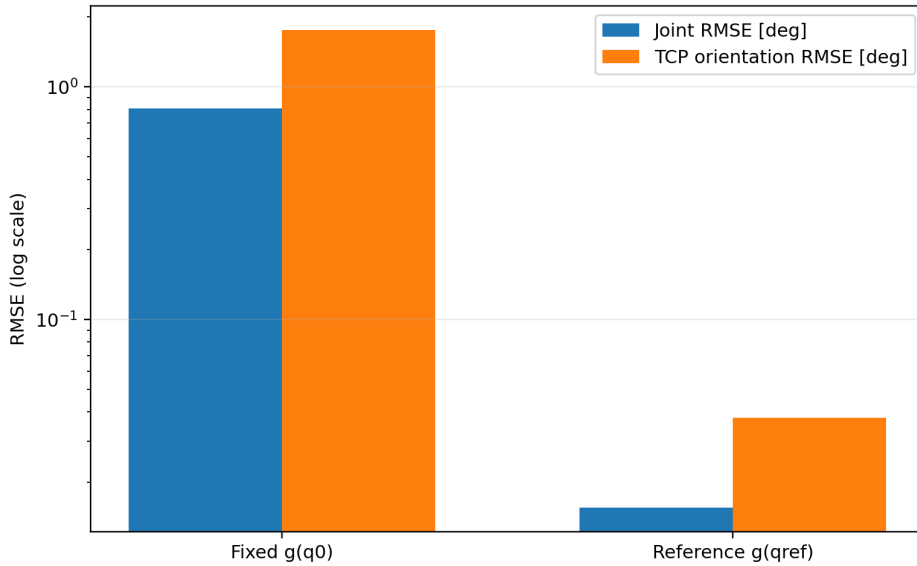


Figure 4: Angular RMSE comparison with identical discrete feedback gain. The logarithmic scale highlights the large reduction obtained by replacing fixed equilibrium gravity with reference-dependent gravity compensation.

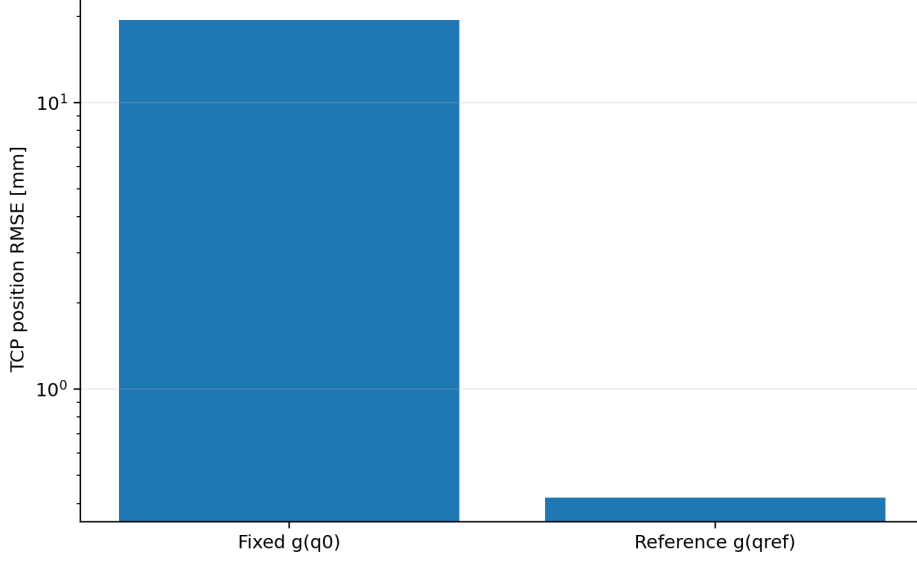


Figure 5: TCP position RMSE for fixed and reference-dependent gravity feedforward. The gain matrix and all other benchmark conditions were unchanged.

The improvement was not purchased through greater actuator usage. Maximum torque utilization decreased slightly from 28.34% to 27.22%, and no saturation occurred in either case. Joint 2 peak torque decreased from approximately 42.52 to 40.83 N m.

This experiment therefore identifies the frozen gravity term as the dominant cause of the poor $\rho_u = 100$ trajectory result. It does not prove that the fixed linear gain is globally valid, nor does it remove all nonlinear model mismatch. It shows that a large predictable configuration-dependent load should not be left entirely to the local feedback term when the gravity model is already available.

9 Final LQR Design

The final nominal controller uses:

- linearization at the benchmark initial equilibrium q_0 ;
- exact ZOH discretization at $T_s = 2$ ms;
- discrete algebraic Riccati equation;
- Bryson-normalized state and torque weights;
- $\rho_u = 100$;
- fixed discrete gain K ;
- reference-dependent gravity feedforward.

The implemented control law is

$$\tau_k = g(q_{ref,k}) - K \begin{bmatrix} q_k - q_{ref,k} \\ \dot{q}_k - \dot{q}_{ref,k} \end{bmatrix}. \quad (33)$$

The final closed-loop discrete spectral radius is

$$0.9900498439. \quad (34)$$

The spectral radius is a property of the fixed local discrete model. It should not be interpreted as a global nonlinear stability certificate.

10 Nominal Benchmark Results

10.1 Joint-space tracking

Table 4: Final discrete LQR joint-space tracking.

Joint	RMSE [deg]	Maximum [deg]	Final [deg]
J1	0.016736	0.029718	-0.014035
J2	0.032212	0.049892	-0.022602
J3	0.005122	0.009038	-0.004278
J4	0.003797	0.007274	0.003255
J5	0.008067	0.017362	0.008439
J6	0.004301	0.007746	-0.003528

The overall joint RMSE is

$$0.015502^\circ. \quad (35)$$

Discrete LQR Joint-Space Trajectory Tracking

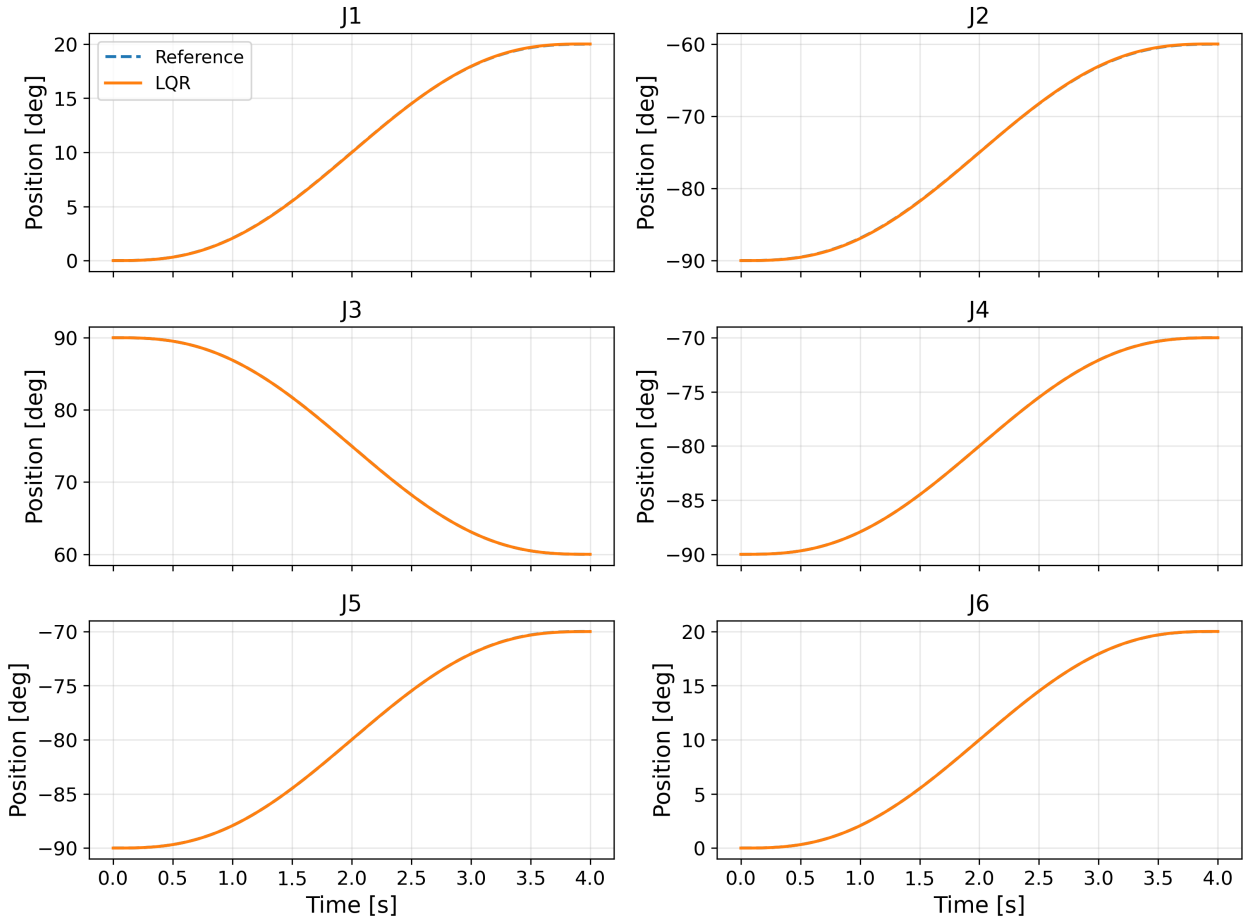


Figure 6: Final discrete LQR joint-space trajectory tracking. Reference and actual trajectories are nearly indistinguishable at the full-motion scale.

Discrete LQR Joint Tracking Error

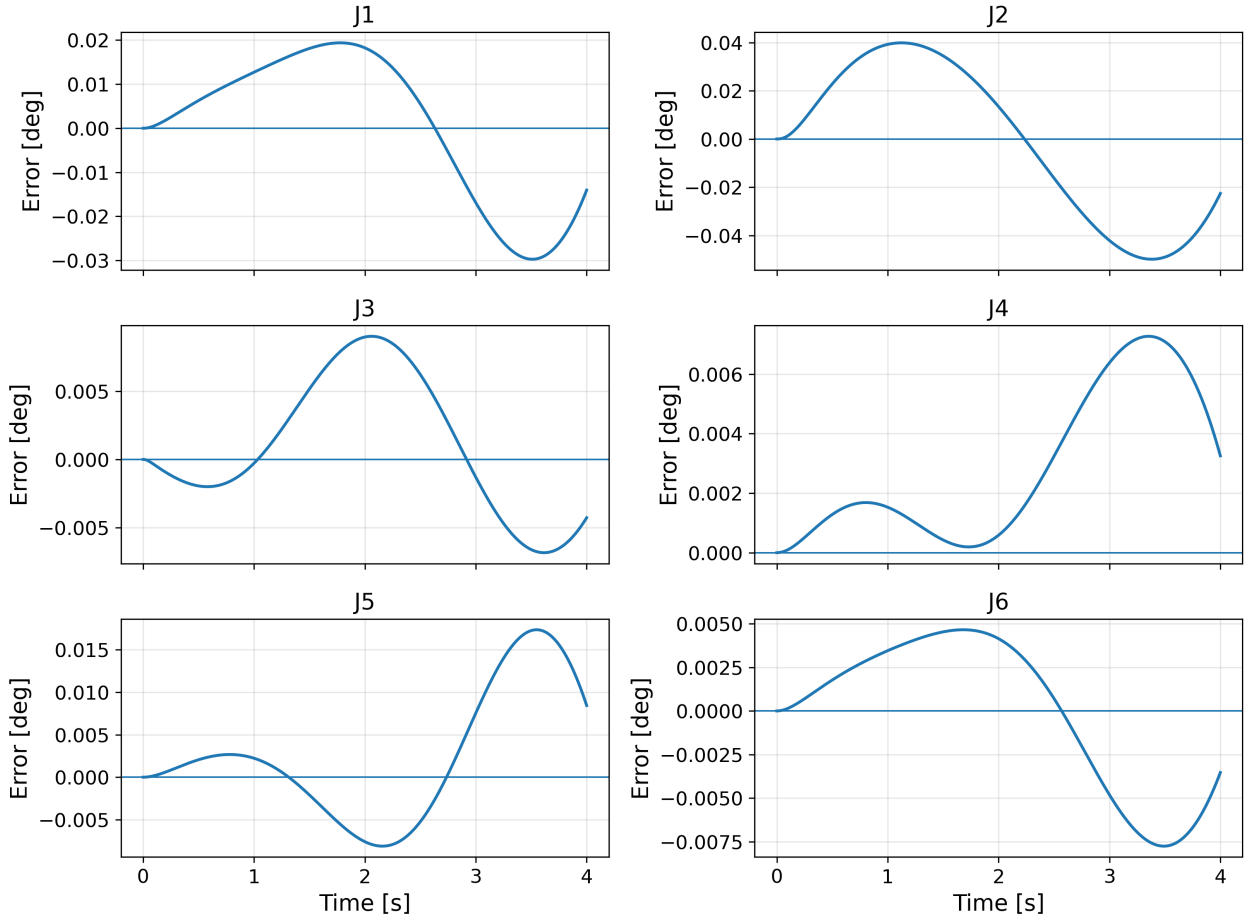


Figure 7: Final discrete LQR joint-space tracking error. Joint 2 exhibits the largest deviation, while the maximum error remains below 0.05° .

Joint 2 has the largest RMSE and maximum error, but its maximum error remains below 0.05° in the nominal benchmark.

10.2 Task-space tracking

Joint-space error was propagated through the same forward-kinematics evaluation used for the other controllers. Position error is the Euclidean translational norm, and orientation error is the geodesic angle of the relative rotation.

Table 5: Final discrete LQR TCP tracking.

Metric	RMSE	Maximum	Final
Position [mm]	0.418511	0.711982	0.337411
Orientation [deg]	0.037763	0.061584	0.029823

The small terminal error and small trajectory-integrated error indicate that the final LQR is not merely reaching the final configuration accurately; it is tracking the path closely throughout the nominal motion.

10.3 Control effort

Table 6: Final discrete LQR control effort.

Joint	Maximum [N m]	RMS [N m]	Utilization [%]
J1	0.209784	0.117172	0.140
J2	40.827162	31.646370	27.218
J3	20.267698	20.127314	13.512
J4	1.825654	1.691061	6.520
J5	0.180134	0.113567	0.643
J6	0.0000127	0.0000071	≈ 0

No torque saturation occurred on any joint.

Discrete LQR Joint Torque Histories

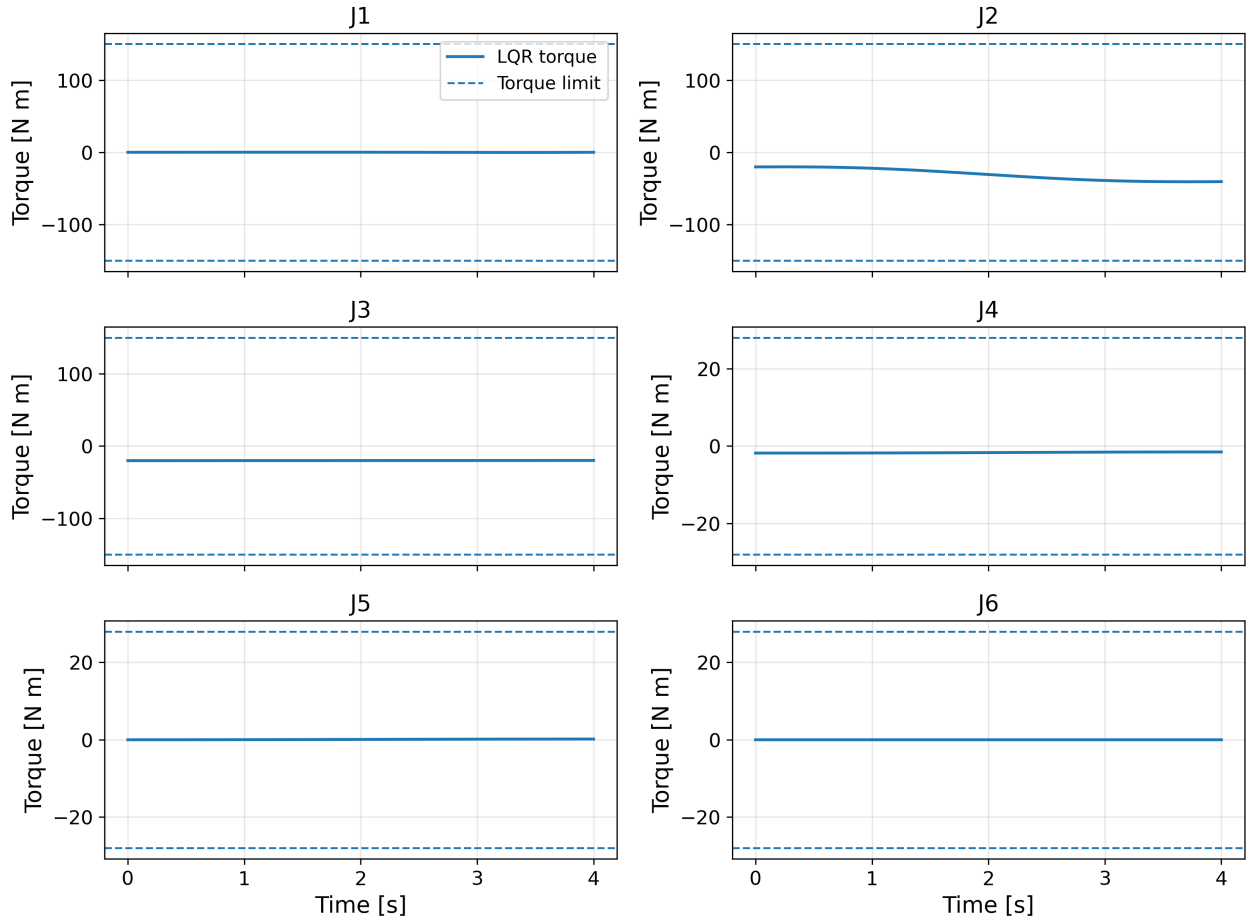


Figure 8: Final discrete LQR joint torque histories and adopted simulation constraints. No joint approaches saturation during the nominal benchmark.

11 Computation Cost

The final timing measurement included the online operations required to evaluate reference-dependent gravity compensation and the LQR state-feedback command. The offline DARE solution and gain construction were excluded.

Measured execution times in the current Python implementation were

$$t_{\text{mean}} = 0.513636 \text{ ms}, \quad (36)$$

$$t_{\text{P99}} = 0.525145 \text{ ms}, \quad (37)$$

$$t_{\text{max}} = 0.913886 \text{ ms}. \quad (38)$$

Relative to the 2 ms control period, these correspond to approximately

$$25.68\% \quad \text{mean}, \quad (39)$$

$$45.69\% \quad \text{measured maximum}. \quad (40)$$

All measured controller samples therefore completed within the nominal 2 ms period in this benchmark environment. This is evidence of computational feasibility for the present Python simulation, not a hard real-time guarantee. It does not include ROS 2 transport, sensor acquisition, operating-system scheduling guarantees, actuator communication, or proprietary UR servo latency.

12 Engineering Interpretation

The study supports several observations.

First, LQR design quality cannot be judged from the continuous-time equilibrium eigenvalues alone. The initial controller stabilized the local continuous linear model, yet failed catastrophically on the sampled nonlinear trajectory. The discrepancy exposed the importance of matching controller design to the implementation sample period and validating the resulting controller on the nonlinear plant.

Second, normalized weighting improved interpretability but did not remove the need for nonlinear validation. Increasing ρ_u reduced the Riccati gain magnitude and relaxed the fastest equivalent modes, but trajectory tracking became substantially worse over the tested nonlinear candidates. Peak torque also did not decrease monotonically. The realized nonlinear response therefore cannot be inferred directly from the numerical magnitude of R .

Third, the feedforward structure was more important than further local gain tuning. With $\rho_u = 100$ fixed, replacing $g(q_0)$ by $g(q_{\text{ref}})$ reduced joint and TCP RMSE by approximately 98%. Because the gain matrix and all benchmark conditions were unchanged, this controlled comparison isolates configuration-dependent gravity compensation as the dominant improvement.

Fourth, the final architecture remains meaningfully different from computed torque. The LQR uses a fixed gain obtained from one local linearization and adds only reference-dependent gravity feedforward. It does not cancel the full nonlinear dynamics through reference acceleration, inertia, and Coriolis/centrifugal terms. The final nominal performance should therefore be interpreted in terms of this specific architecture rather than as a generic property of “LQR.”

Finally, nominal accuracy is not robustness. The present controller is evaluated against the same nominal model used to construct its linearization and gravity term. Payload, parameter mismatch, disturbance, and other modeling errors are reserved for the common robustness study after all controller families have been implemented.

13 Relation to the PID and Computed-Torque Baselines

The common nominal benchmark gives the following reference values:

Table 7: Nominal benchmark context for the controllers completed to this stage. Computation times are implementation-dependent and are reported only as measured in the current Python study.

Controller	Joint RMSE [deg]	TCP RMSE [mm]	Mean computation [ms]
PID	0.4698	5.836	0.0247
Discrete LQR + gravity FF	0.015502	0.418511	0.5136
Computed torque	0.001244	0.004139	8.1597

The table is not intended as an overall ranking. The three controllers use different model information and impose different computational burdens. PID is predominantly reactive feedback, the final LQR combines fixed optimal state feedback with configuration-dependent gravity compensation, and computed torque evaluates the nonlinear dynamics directly. A broader assessment requires the common robustness and implementation-complexity study rather than nominal tracking error alone.

14 Limitations

- The plant is an ideal fixed-base rigid-body UR5e model with no external payload.
- Friction, backlash, compliance, actuator electrical/gearbox dynamics, sensor noise, communication delay, and the proprietary UR servo are not modeled.
- The adopted torque values are simulation constraints carried from the imported model, not independently verified physical motor torque ratings.
- The LQR gain is fixed from a single linearization at q_0 . No gain scheduling or time-varying Riccati solution is used.
- Reference-dependent gravity compensation uses the nominal model. Its performance under payload or parameter mismatch has not yet been established.
- The feedforward term contains gravity only. Desired acceleration and Coriolis/centrifugal feedforward are not included in the final LQR architecture.
- The Bryson “allowable” state values are normalization scales, not hard constraints.
- The selected $\rho_u = 100$ is the best performer within the predefined nonlinear candidate range tested here; it is not claimed to be a global optimum.
- The discrete spectral radius characterizes the local linear model and is not a global nonlinear stability proof.
- Controller timing is implementation- and hardware-dependent. The reported Python timing is used as an engineering benchmark, not a real-time certification.
- Robustness to payload, parameter mismatch, and external disturbance is intentionally deferred to the common multi-controller comparison. The common study will apply identical payload/model-mismatch and external-disturbance conditions to PID, computed torque, LQR, and MPC; perturbation magnitudes will be defined after the nominal MPC benchmark is completed.

15 Conclusion and Transition

The LQR study began with a controller that was mathematically stable around the initial continuous-time equilibrium but unusable on the sampled nonlinear trajectory. Rather than hiding that failure, I used it to separate three design questions: sampled-data formulation, state/input weighting, and nonlinear feedforward structure.

Reformulating the controller in discrete time at the actual 2 ms sample period eliminated the continuous-versus-discrete design mismatch. Bryson normalization then provided interpretable state and torque scaling, while the ρ_u study established a practical feedback tradeoff within the tested design range. The remaining tracking error was finally traced to the use of fixed equilibrium gravity compensation.

Changing only the feedforward term from $g(q_0)$ to $g(q_{\text{ref}})$ reduced overall joint RMSE from 0.810459° to 0.015502° and TCP position RMSE from 19.474 to 0.419 mm without increasing torque utilization or causing saturation. This result is the central engineering finding of the LQR stage: a

well-designed local feedback law can still perform poorly if a large predictable configuration-dependent load is represented incorrectly.

The final nominal LQR therefore combines a fixed discrete optimal state-feedback gain with reference-dependent gravity compensation. It provides a substantially more accurate nominal trajectory baseline than the feedback-only PID while remaining computationally feasible in the present Python benchmark.

The next controller stage is model predictive control. MPC will retain a model-based state-space formulation but introduce explicit finite-horizon prediction and constraint handling. After MPC is complete, PID, computed torque, LQR, and MPC will be evaluated under common robustness conditions so that nominal accuracy, model dependence, computation, constraint handling, and performance degradation can be assessed together.